Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 6

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : both
Restarts   : 20

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

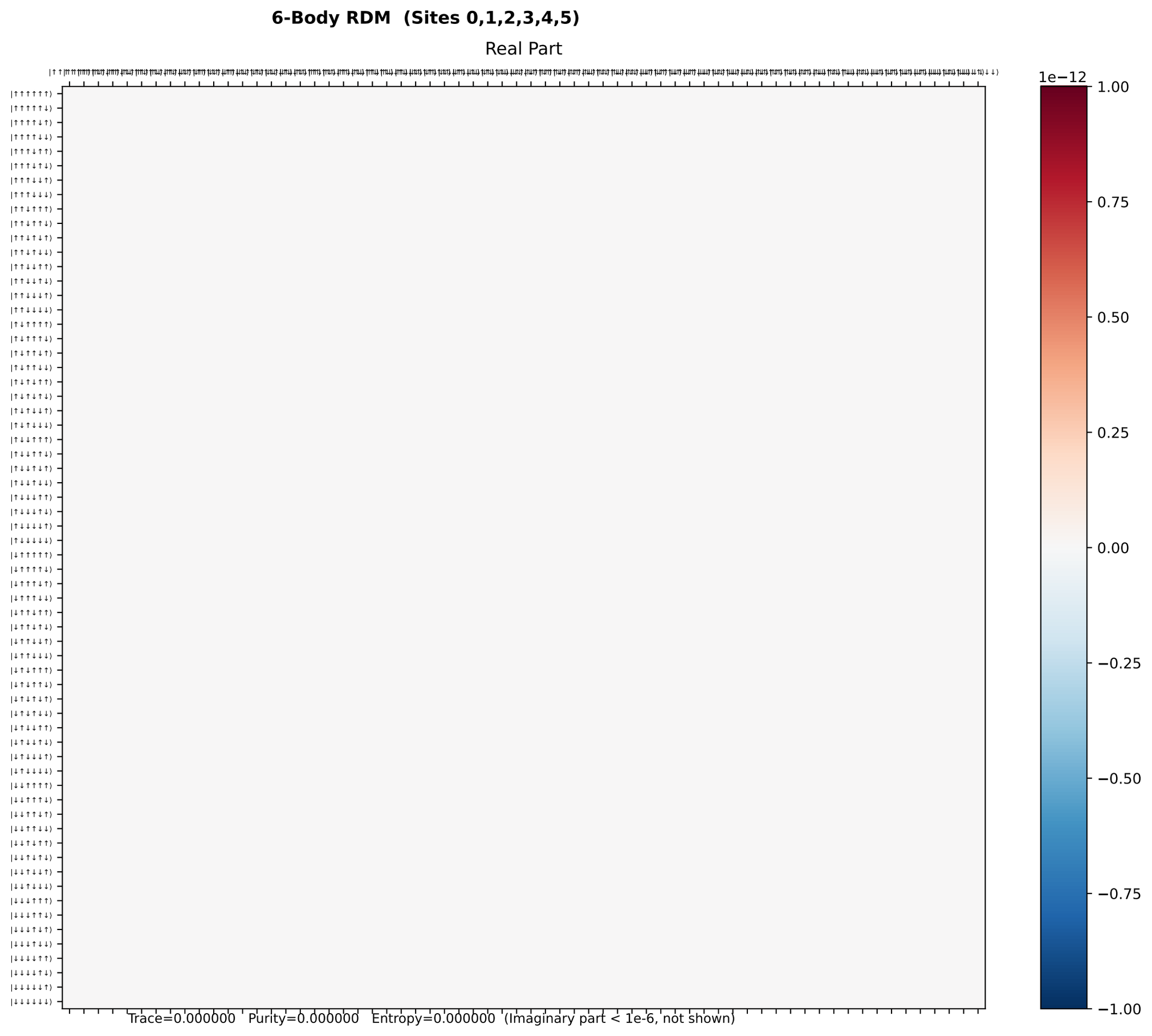
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Body order : 6
Clusters   : 1

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Summary — 6-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_expo	C_stiefel	C_UB	State
0,1,2,3,4,5	0.000000	—	0.000000	0.000000	0.000000	0.000000	STRONGLY MIXED


6-Body RDM (Sites 0,1,2,3,4,5)
Real Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Trace=0.000000 Purity=0.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 0,1,2,3,4,5)

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RDM Metrics
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  Trace(rho)           : 0.0000000000
  Purity Tr(rho^2)     : 0.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -4.442239e-44
  Eigenvalue max       : 1.539833e-28
  Imaginary part       : < 1e-6 (not shown)
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GME Analysis
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  gamma = Tr(rho^2) = 0.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
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Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
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Convex Roof Analysis
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  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = nan   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.00000000   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.00000000

  Status: no_GME
  Bounds consistent: CHECK
  Convergence: TIGHT
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